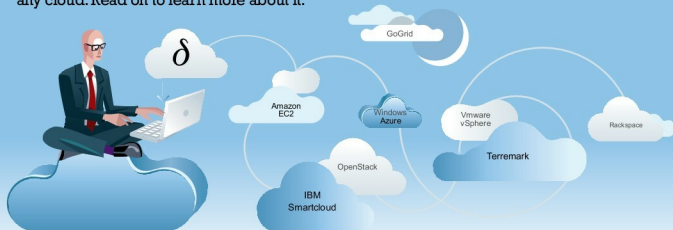


Manage Multiple Clouds with Apache Deltacloud

Apache Deltacloud provides a unified API that can be used to manage services on any cloud. Read on to learn more about it.



Cloud computing has grown tremendously over the last couple of years. Organisations across industries have realised the benefits of using the cloud environment. Cloud computing has three service models: 1) Software as a Service, 2) Platform as a Service, and 3) Infrastructure as a Service. In turn, these cover four deployment models: 1) the public cloud, 2) the private cloud, 3) the hybrid cloud, and 4) the community cloud. In recent years, the vendor landscape of cloud services across service and deployment models has grown exponentially.

There are many cloud service providers, which has resulted in many complexities. Most providers offer their own set of APIs such as AWS, Eucalyptus, OpenStack, VMware vCloud, etc. Some APIs support multiple cloud environments or have many similarities with other APIs but still, some basic concerns are not addressed. A standard API can help to address one of the main concerns—vendor lock-in. A standard API can be very useful in managing multiple clouds from a single Web UI. It is easy to create a customised application for common use cases such as scaling of infrastructure across multiple cloud environments. Hence, it is useful to have one API to manage multiple cloud environments. Deltacloud is an API developed by the Apache Software Foundation and Red Hat. It abstracts the differences between Infrastructure as a Service implementations.

An overview of Deltacloud

The Deltacloud project was started by Red Hat in 2009. The founders identified a big gap in terms of portability

and data lock-in. Writing applications for any development or deployment with cloud service providers resulted in a locked mode situation with the cloud vendor's API. Hence, Red Hat wanted to offer a single API, developed openly, for multiple clouds.

The Deltacloud API incubated under the Apache Foundation (ASF) is a REST-based cloud abstraction API to manage resources in different Infrastructure as a Service clouds. Motivation to build a standard API for multiple clouds was to have a heterogeneous cloud and virtualisation environment. Since it is a REST-based API, it is programming language agnostic. Deltacloud supports HTML, XML and JSON. In the absence of a unified cloud API, Deltacloud is the next best thing. Similar to Deltacloud, there are other open source projects to develop cloud abstraction APIs such as jclouds and libcloud. The former is specific to Java, while libcloud is specific to Python. Deltacloud is a very good example of how open source initiatives continue to drive innovation and can become game changers.

Features of the Deltacloud API

- A basic set of CRUD operations
 - Create (instance)
 - Read/List (instances, images, hardware profiles...)
 - Update (instance state—reboot, stop, start)
 - Destroy (instance, images)
- Extended operations
 - Upload blob to bucket (S3)
 - Instance authentication